

Microcontroller Project Proposals — Full Review

For: Chamidya, De Silva, Omesh, Kalansooriya, Goonatillaka Covers: 3D Potter Printer · Hydro-Dipping Workstation · AquaScout ROV · Autonomous Ring Toss Launcher

How to use this document

Each proposal gets four sections:

1. **Requirement check** — pass / partial / fail against your supervisor's stated criteria, with the specific gap named.
2. **Supervisor Q&A prep** — draft answers to the seven questions you listed, written the way you'd actually say them in the viva.
3. **Component research** — every sensor and the MCU get the full Purpose/Accuracy/Cost/Alternatives/Problems (or Pros/Cons) treatment, since those are what get cross-examined. Structural/mechanical/wiring items are grouped in a shorter table — a supervisor rarely interrogates why you chose M3 bolts.
4. **Approval checklist** — strict pass/fail against your General / Hardware / Proposal criteria.

Section 6 ranks the four against each other. Section 7 is the work-distribution fairness review you asked for, across all four decks.

All costs are in LKR and are **indicative** — sourced from typical hobbyist/industrial component pricing, not a live quote. Verify against a local supplier (e.g. Otronix, RoboMart-type distributors, or direct AliExpress/Digikey/Mouser imports for the industrial-grade parts) before locking the budget slide.

1. Quick-Look: Requirement Fulfillment Across All Four

Requirement	3D Potter Printer	Hydro-Dipping Workstation	AquaScout ROV	Ring Toss Launcher
Solves a real-world problem	Partial — automates a craft, niche market	Strong — real safety/consistency problem in a real industry	Partial — real problem, but heavily "solved" already by hobbyist ROV projects	Weak — a game, not a problem
New idea / meaningful improvement	Partial — dual-mode (pottery+sculpting) is the genuine novelty	Good — safety/monitoring layer is the novel part	Weak — closely follows published DIY ROV designs (cited directly)	Good — vision-guided aiming is a real differentiator over toy launchers
Hardware-focused	Yes	Yes	Yes	Borderline — Pi 5 + YOLOv8 is software/AI-heavy
AI only as value-add	Yes (none used, fine)	Yes (none used, fine)	Yes (none used, fine)	Risk — CV/YOLO is central to function, not a bonus
Low budget	~87,000–95,000 LKR	~112,200 LKR	~97,750 LKR	156,420 LKR — highest by far
Fully practical (not just prototype)	Yes, if drying/firing steps are scoped honestly	Yes	Yes, with caveats on depth rating	Yes, but "practical use" is limited to demos/competitions
Testable in lab	Yes	Yes	Needs a water tank/pool — arrange in advance	Yes, needs an open/safe indoor space

Requirement	3D Potter Printer	Hydro-Dipping Workstation	AquaScout ROV	Ring Toss Launcher
5 members, ≥1 sensor each	Yes (see 7.1)	Yes, one assignment is weak (see 7.2)	Yes, one sensor invented late (see 7.3)	Yes (see 7.4)
Inputs/outputs clearly defined	Yes	Yes	Yes	Yes
Scope clearly defined	Yes	Yes	Yes	Mostly — "consistent landing" needs a numeric target

Bottom line: Hydro-Dipping is currently your strongest proposal on "real-world problem" and safety engineering. Ring Toss is your weakest on "real-world problem" and budget, and carries an AI-centrality risk your supervisor will likely flag first. AquaScout's biggest exposure is novelty — it's a well-known project type with public build guides, so you need to be crisp about what's actually new. 3D Potter Printer is solid but should tighten what "fully practical" means (is bisque firing outside the ceramic itself in scope, or not?).

2. Project A — 3D Potter Printer

2.1 Requirement Check

Criterion	Status	Notes
Real-world problem	 Partial	Automating hand-thrown pottery is real but niche (craft studios, small-batch ceramics, education). Frame it as "consistency + accessibility for non-potters," not "replacing potters."
New/improved		Dual-mode (rotary pottery + XYZ sculpting on one frame) is a genuine differentiator vs. single-mode desktop clay printers.
Hardware-focused		Motion, extrusion, mixing, drying are all physical subsystems.
AI as value-add only		No AI claimed — fine, don't force it in.
Low budget		~87,000–95,000 LKR is reasonable for 3-axis motion + extrusion.
Fully practical	 Partial	Firing is explicitly pushed to "external" (slide 7, step 11) — make sure your scope slide says the printer produces greenware/decorated bisque-ready pieces , not fired ceramics, so nobody assumes you're building a kiln too.
Testable in lab		Motion, extrusion, moisture control are all bench-testable without firing.
5 members / ≥1 sensor each		See §7.1 — confirmed, one sensor each.
Inputs/outputs defined		Clay+water+G-code in; printed/decorated object out.
Scope defined	 Mostly	Tighten: is "bust/sculpture" mode fully in scope for Level 01, or a stretch goal? A 3-axis gantry doing wet-clay sculpture is mechanically harder than pottery mode — be ready to say which is the guaranteed deliverable and which is a stretch.

2.2 Supervisor Q&A Prep

Why is this better than existing products? Commercial clay 3D printers (e.g., WASP, Delta WASP clay kits) cost well into six figures LKR-equivalent and are single-mode (either pottery or sculpting, rarely both). Your dual-mode frame — rotary bed for pottery, XYZ gantry for sculpture/bust work, on shared hardware — is the cost/versatility argument. Also

lead with the closed-loop material prep (moisture + load-cell feedback mixing) — most desktop clay printers assume pre-mixed clay is loaded manually; yours mixes and validates consistency automatically before it ever reaches the extruder.

Why did you choose this sensor? (per-sensor answers are in §2.3 below — memorize the one-line "why" for each)

What happens if this sensor fails?

- Moisture sensor fails → extrusion continues blind; you should add a **fallback**: if load-cell resistance readings drift outside the calibrated band for >N seconds with no moisture reading, auto-pause and flag "check material" rather than trusting one signal. Currently this failure mode isn't addressed in the flow — add it before the viva.
- Limit switch fails → axis can't home; over-travel risk. Firmware should treat "no switch trigger within expected travel distance" as a hard fault, not just rely on the switch existing.
- Load cell fails/drifts (common with cheap HX711 boards) → extrusion pressure control is blind; pair it with a simple current-draw sanity check on the extruder motor as a secondary signal (you already have this — motor current sensor — so say explicitly that it's your load-cell failure backup).

Why not use another microcontroller? ESP32 over Arduino Mega: dual-core (one core can run motion/PID while the other handles UI/logging), built-in Wi-Fi if you ever want status logging, faster clock for smoother stepper pulse generation, and it's cheaper. Arduino Mega is a fallback if you need more physical UART/PWM pins than the ESP32 breaks out easily, but for this design ESP32 is the better fit.

Estimated cost? Rs. 87,000–95,000 (slide 15) — walk in ready to justify the two biggest lines: frame (18,000) and motors (16,000), since those are 39% of the budget combined.

Power consumption? 24V/10A SMPS (~240W) sized for motors+electronics; you have a runtime calc for the battery backup (~1.5 hrs on a 240Wh pack) — know that this backup is for **safe shutdown**, not continued printing, and say so proactively.

How will you test it?

- Axis accuracy: repeatability test (home → move to fixed point → measure deviation with calipers, N repeats).
- Extrusion consistency: fixed-time extrusion at a set pressure, weigh output, check variance across trials.
- Moisture control loop: deliberately load over-wet and over-dry clay, confirm the mix→measure→adjust loop converges.
- Full print: run a simple test geometry (cylinder) end-to-end and inspect dimensional accuracy and surface consistency.

2.3 Component Research

Microcontroller — ESP32 DevKit V1

Pros: Dual-core (motion control + UI/logging in parallel), built-in Wi-Fi/BT for future status dashboard, plenty of PWM-capable pins for 4–5 stepper drivers, cheap, huge community support for stepper/G-code libraries. **Cons:** Fewer native hardware UARTs than a Mega; 3.3V logic needs level-shifting if any sensor/module is 5V-only (check your HX711 and DHT22 board variants); real-time determinism is weaker than a dedicated motion controller if you push many axes at high step rates. **Cost:** ~Rs. 5,500 (matches your budget line). **Supplier:** Local robotics/electronics retailers or AliExpress/Digikey for the genuine DevKit V1. **Reason to choose:** Best balance of processing headroom, pin count, and cost for a 3–4 axis stepper system with sensor fusion and a UI.

Sensor 1 — Moisture Sensor (clay water content)

Purpose: Detect 25–35% water content in the clay mix; prevents nozzle clogging, gates the mix→measure→adjust loop. **Accuracy:** Capacitive soil-moisture-style sensors used for clay are typically $\pm 3\text{--}5\%$ relative — good enough for a "dry/OK/wet" band decision, not for scientific measurement. State that explicitly; don't oversell precision. **Cost:** ~Rs. 300–800 depending on probe type (capacitive vs. resistive). **Alternatives:** Resistive probes (cheaper, corrode faster in wet clay — a real durability concern you should mention), or infer moisture indirectly from load-cell/extrusion-resistance only (your backup signal). **Problems:** Capacitive probes drift with clay mineral content and temperature; resistive probes corrode with repeated water contact. Plan to re-calibrate the sensor whenever clay type changes.

Sensor 2 — Load Cell (5–20 kg) + HX711

Purpose: Measures extrusion resistance/pressure to detect clay that's too thick (high resistance) or too runny (low resistance). **Accuracy:** HX711 is a 24-bit ADC — resolution is excellent, but real-world accuracy is limited by load-cell quality and mechanical mounting; expect $\pm 2\text{--}3\%$ of full scale with a budget cell. **Cost:** ~Rs. 800–1,500 for load cell + HX711 breakout combined. **Alternatives:** Strain-gauge pressure sensor directly in the extrusion chamber (more expensive, more direct measurement, harder to mount); motor current alone (cheaper, less direct, more noise). **Problems:** HX711 modules are notoriously noisy without proper shielding/averaging; needs software filtering (moving average) or readings will be jittery enough to false-trigger your adjust loop.

Sensor 3 — Ultrasonic Sensor (nozzle standoff height)

Purpose: Keeps consistent nozzle-to-surface distance for uniform layer thickness. **Accuracy:** Standard ultrasonic (e.g. HC-SR04 class) gives ~2–3mm resolution at close range — adequate for layer-height control on a clay printer (layers are usually several mm, unlike FDM plastic printing). **Cost:** ~Rs. 300–500. **Alternatives:** IR distance sensor (worse on wet, uneven clay surfaces which scatter IR poorly); mechanical feeler/limit switch per

layer (simpler, slower, no continuous feedback). **Problems:** Ultrasonic readings get unreliable on very wet, glossy, or highly irregular clay surfaces (sound scatter); mount rigidly and filter noisy readings.

Sensor 4 — DHT22 (Temperature & Humidity)

Purpose: Monitors ambient conditions for drying-rate management and preventing cracking. **Accuracy:** $\pm 0.5^{\circ}\text{C}$, $\pm 2\text{--}5\%$ RH — perfectly adequate for environmental monitoring (not a precision requirement here). **Cost:** ~Rs. 400–600. **Alternatives:** SHT31 (better accuracy and faster response, slightly pricier, ~Rs. 800–1,200) if drying-crack control becomes a bigger part of your evaluation criteria. **Problems:** Slow sample rate (~ 0.5 Hz) — fine for drying monitoring, would be a problem if you tried to use it for anything faster.

Sensor 5 — Limit Switches $\times 3$ (X, Y, Z homing)

Purpose: Establishes home position per axis; essential for repeatable auto-bed leveling. **Accuracy:** Mechanical switches are effectively binary/exact at the trigger point (sub-mm repeatability with a good microswitch) — this is a strength, not a weakness. **Cost:** ~Rs. 300–500 for a set of 3. **Alternatives:** Optical endstops (more precise, more expensive, dust-sensitive — a real concern given clay dust/splatter); Hall-effect sensors with a magnet (cleaner, no physical wear, slightly pricier). **Problems:** Mechanical switches wear out over thousands of cycles and can get clogged with clay residue — plan an enclosure/wiper for them.

Bonus sensor mentioned only in team-breakdown slide — Motor Current Sensor (INA219)

This appears in the Hardware Components table under "Smart Consistency Detection" but is assigned as a sensor responsibility on the team-breakdown slide — make sure both slides agree on what it does (secondary check on extrusion torque, and general motor-health monitoring). **Purpose:** Detects abnormal current draw on the mixer/extruder motors — indicates jams, over-thick clay, or mechanical binding. **Accuracy:** INA219 current sensing is accurate to a few mA — plenty of resolution for a "normal vs. abnormal load" decision. **Cost:** ~Rs. 400–600. **Alternatives:** ACS712 (cheaper, analog, noisier); relying on load cell alone (loses the independent cross-check you get from having both). **Problems:** Needs correct current range/shunt selection for your motor's actual draw, or resolution suffers.

Actuators & Structural Components (grouped)

Component	Function	Est. Cost	Reason to choose
NEMA 17 stepper ×3 (X/Y/Z) + TMC2208/A4988 drivers	3-axis motion	~Rs. 16,000 (motors) + included in drivers line	Standard, well-documented, cheap, silent with TMC2208
High-torque DC gear motor + auger	Clay mixing	included in extruder line	Gear motor gives torque needed for wet clay without stalling
NEMA 17 (extruder)	Auger/piston extrusion drive	included above	Matches motion-system parts, one spares pool
Mini 12/24V water pump	Auto water dosing	~Rs. 500–1,000	Simple, cheap, closes the mixing loop without manual water addition
20×4 LCD or TFT + keypad/SD	User interface & file input	~Rs. 4,000	Low cost, no dependency on a phone/PC during printing
Aluminium 2020/2040 frame	Structural rigidity	~Rs. 18,000	Standard 3D-printer-grade extrusion, easy to source and assemble
24V SMPS + battery backup	Power + safe- shutdown reserve	~Rs. 6,000 (PSU)	Battery backup specifically protects an in-progress print from a power cut — a nice real- world justification point

2.4 Approval Checklist — 3D Potter Printer

General

- ☒ Solves a real problem — ⚠️ frame narrowly (craft consistency/accessibility, not "replacing potters")
- ☒ Target users identified — should be made explicit on a slide (hobbyists, small studios, education) — currently implied, not stated
- ☐ Existing solutions researched — **missing**: no competitor/benchmark slide or reference list (unlike the other three decks, this one has none). Add a references slide and 1–2 lines comparing to commercial clay printers.
- ☒ Improvement clearly explained — dual-mode + closed-loop material prep

Hardware

- ✓ Hardware-focused
- ✓ Enough meaningful sensors — 6 sensors across 5 members, each contributing to a real control decision
- ✓ Sensors improve functionality — yes, each ties to a specific failure mode (clog, drift, misalignment)
- ✓ Components available — all standard hobbyist parts
- ✓ Low power — reasonable for a benchtop system
- ✓ Low cost — ~90,000 LKR, comparable to the others
- ✓ Testable in lab — yes, without firing
- ✓ Can be built — yes, standard 3-axis gantry construction

Proposal

- ✓ Inputs identified
- ✓ Outputs identified
- ✓ Block diagram complete
- ✓ Flowchart complete (operational flow, 12 steps)
- ✓ Component explanations ready — after this document, yes
- ☐ Scope defined — **tighten:** is bust/sculpture mode guaranteed or stretch? Is firing in/out of scope? State both explicitly.
- ✓ Budget estimated

Verdict: Approve with revisions — add a competitor/reference slide, and add one explicit scope-boundary sentence (firing excluded; sculpture mode may be a stretch goal).

3. Project B — Hydro-Dipping Workstation

3.1 Requirement Check

Criterion	Status	Notes
Real-world problem	 Strong	Manual hydro-dipping has documented chemical-exposure and consistency problems in small automotive/product-customization shops — this is your most defensible "real problem" claim of the four.
New/improved		The novelty isn't the dipping itself (established process) — it's the safety/monitoring layer (VOC gating, leak detection, enclosed ventilation) that hobbyist/small-shop setups genuinely lack. Say this explicitly.
Hardware-focused		
AI as value-add only		None claimed.
Low budget		112,200 LKR — a bit higher than the others, driven by the enclosure (15,000) and SMPS units (7,500) — justifiable given the safety requirements.
Fully practical		Unlike the printer's firing question, this system's full process (film→spray→dip→dry) is genuinely self-contained.
Testable in lab		Every stage is bench-testable; VOC/leak/temp sensors can all be validated with known stimuli.
5 members / ≥1 sensor each	 Partial	See §7.2 — one member's "sensor" assignment is UI-related, not a real sensor.
Inputs/outputs defined		
Scope defined		Clear 9-step operational flow with a worked example cycle — this is your best-specified proposal.

3.2 Supervisor Q&A Prep

Why is this better than existing products? Small-scale hydro-dipping today is almost entirely manual and open-air — no containment, no monitoring, and full operator exposure to activator fumes. Commercial industrial hydrographic lines exist, but at a scale and cost

far beyond a small shop or educational lab. Your system sits in the gap: compact, enclosed, and sensor-monitored, at hobbyist-adjacent cost.

Why did you choose this sensor? (see §3.3)

What happens if this sensor fails?

- VOC sensor fails/reads low → spray could continue past a safe fume threshold. This is your single biggest safety-critical failure mode — say clearly that the design should **fail safe**: if the VOC sensor gives no valid reading (not just a high reading) within its expected refresh interval, the system should halt the spray and refuse to unlock the enclosure, not assume "no signal = safe."
- Leak sensor fails → water/chemical leak undetected. Pair with a simple redundant float switch or drip-tray moisture check as a cheap secondary safety net — currently the design relies on one capacitive sensor for a safety-critical function.
- Position limit switch fails → dipping rail over-travels. Same firmware principle as the printer: treat "no trigger within expected travel" as a fault.

Why not use another microcontroller? ESP32 industrial dev board over a plain Arduino: dual-core lets you run FreeRTOS tasks — one core for safety-critical sensor polling (VOC, leak, E-stop) running at guaranteed intervals, the other for UI/logging — which matters more here than in the other projects because this system has genuine safety-interlock requirements.

Estimated cost? LKR 112,200 total (with 10% contingency already built in) — be ready to explain the two priciest lines: enclosure/frame (15,000) and the three SMPS units (7,500), both justified by the safety/containment requirement, not scope creep.

Power consumption? Three regulated rails (24V motors/fans/solenoid, 12V spray, 5V logic) off a protected AC mains input with fuse/MCB/EMI filter/E-stop relay — know your protection chain by name, this is exactly what a supervisor will probe on a chemical-adjacent system.

How will you test it?

- VOC threshold test: introduce a known low-concentration solvent source, confirm spray halts at threshold.
- Leak test: deliberately introduce a small controlled water leak, confirm detection and safety cutoff.
- Dipping repeatability: run N cycles, measure position consistency via encoder/limit-switch logs.
- Full cycle: run the worked example (film feed → spray → dip → dry) end-to-end and time each stage against spec.

3.3 Component Research

Microcontroller — ESP32 Industrial Development Board

Pros: Dual-core for safety-critical task separation (sensor polling vs. UI), Wi-Fi/BT for the optional dashboard, industrial-rated boards typically add better power-supply tolerance and connector robustness than hobby boards. **Cons:** Industrial variant costs more than a plain DevKit; still 3.3V logic, so integrating any 5V industrial sensor module needs a level shifter. **Cost:** ~Rs. 3,500 (matches your budget). **Supplier:** Industrial-automation distributors or Espressif-authorized resellers; hobby DevKits from local electronics stores are a cheaper fallback if the "industrial" rating isn't strictly required for the lab demo. **Reason to choose:** Matches the "industrial thinking" positioning of the whole proposal and gives genuine safety-task isolation via dual-core FreeRTOS.

Sensor 1 — PT100 RTD + MAX31865 (Water Temperature)

Purpose: Confirms water is within the process's required temperature band before spraying/dipping. **Accuracy:** PT100 + MAX31865 gives $\sim \pm 0.1\text{--}0.5^{\circ}\text{C}$ — genuinely industrial-grade precision, notably better than a thermistor or DS18B20. **Cost:** ~Rs. 4,300 combined (2,500 + 1,800) — one of your pricier single measurements. **Alternatives:** DS18B20 digital thermistor (~Rs. 300, far cheaper, $\pm 0.5^{\circ}\text{C}$ typical but drifts more over time and range) — a fair "why not the cheap one" answer is that PT100 is the industry-standard choice for stable long-term accuracy, which matters because water temperature affects both film activation and coating quality. **Problems:** SPI wiring to MAX31865 needs careful reference-resistor selection matching the specific PT100 (2/3/4-wire) — a common integration mistake, worth mentioning you've accounted for it.

Sensor 2 — Industrial Ultrasonic Sensor (Water Level)

Purpose: Confirms adequate water level before film placement/dipping. **Accuracy:** Industrial ultrasonic level sensors typically resolve to a few mm over their rated range — plenty for a tank-level go/no-go decision. **Cost:** ~Rs. 6,500 — notably pricier than a hobby HC-SR04; justify by needing reliable operation near a water surface with foam/ripple, which cheap modules handle poorly. **Alternatives:** Float switch (cheap, binary only, no continuous level read); capacitive level sensor (cheaper than industrial ultrasonic, less proven for open-water surface conditions). **Problems:** Surface ripples/foam from the activator spray can cause false readings — mount away from direct spray path and consider averaging over several readings.

Sensor 3 — SGP30 / ENS160 (VOC / Air Quality)

Purpose: Detects activator fume concentration; the core safety sensor of the whole system — gates spraying and enclosure unlock. **Accuracy:** These are indoor-air-quality-grade sensors (ppm-level VOC index, not lab-grade gas chromatography) — accurate enough for a threshold/alarm decision, **not** for regulatory-grade exposure measurement. State this limitation clearly; don't claim it as a certified safety instrument. **Cost:** ~Rs. 3,200.

Alternatives: MQ-series gas sensors (cheaper, less specific, more drift, need burn-in); a dedicated solvent-specific electrochemical sensor (far more accurate for the specific activator chemical, but much pricier and harder to source). **Problems:** These sensors need a warm-up/baseline period and drift over their lifetime — plan periodic recalibration, and treat a sensor fault as "assume unsafe," not "assume clear" (see Q&A above).

Sensor 4 — Omron Industrial Limit Switches ×4 (Position Detection)

Purpose: Defines travel limits for the dipping rail. **Accuracy:** Mechanical, effectively exact at the trigger point. **Cost:** ~Rs. 3,800 for 4 — pricier than generic microswitches because Omron units are rated for repeated industrial-duty cycling. **Alternatives:** Generic microswitches (cheaper, shorter service life); optical/magnetic proximity sensors (contactless, no mechanical wear, higher cost). **Problems:** Splash/moisture near a water tank is a real concern for any switch — confirm IP rating is adequate for the mounting location.

Sensor 5 — Capacitive Liquid Leak Sensor ×2

Purpose: Detects water/chemical leaks inside the enclosure — a genuine safety function. **Accuracy:** Binary presence/absence detection, essentially reliable at the point of contact — the limitation is coverage area, not measurement precision. **Cost:** ~Rs. 3,000 for 2. **Alternatives:** Simple conductive-probe leak sensor (cheaper, can corrode/foul over time with chemical exposure); float switch in a drip tray (simpler, very reliable, worth adding as a redundant backup given this is safety-critical). **Problems:** Coverage is local to sensor placement — a leak elsewhere in the enclosure won't be caught. Consider 1 more unit or a drip-tray placement strategy and say so.

Sensor 6 — Hall-Effect Airflow Sensor

Purpose: Confirms the exhaust fan is actually moving air (not just powered on) before allowing drying/unlock. **Accuracy:** Good for a binary "airflow present/absent" decision; not a precise CFM measurement in this configuration. **Cost:** ~Rs. 2,200. **Alternatives:** Simple fan-tachometer feedback (cheaper, confirms fan is spinning but not that air is actually flowing through a possibly-blocked duct — airflow sensor is the better safety check); thermal anemometer (more accurate, much pricier, overkill here). **Problems:** Needs correct placement in the duct to avoid false positives from turbulence.

"Sensor 7" — User Input Interface Sensing (assigned to Goonatillaka)

This is the one questionable "sensor" on the team-breakdown slide — flagged in §7.2. It's really an input device (LCD touchscreen/rotary encoder), not a sensor in the same sense as the other six. Recommend re-labeling this member's assigned sensor as a **real physical sensor** — e.g., co-own the second leak sensor, or make them the owner of a to-be-added redundant safety float switch (see Sensor 5 problem above) — so all five members are genuinely defending a sensor choice in the viva.

Actuators & Structural Components (grouped)

Component	Function	Est. Cost	Reason to choose
NEMA 17 + TMC2209 + MGN12 rail	Precision dipping motion	~Rs. 3,000 + 1,600 + 7,500	TMC2209 is near-silent, MGN12 rail gives smooth, low-play linear travel
Solenoid valve + HVLP nozzle	Activator spray	~Rs. 2,800 + 2,000	Solenoid gives precise, repeatable spray timing vs. manual triggering
Brushless centrifugal fans ×2 + carbon filter	Ventilation & fume extraction	~Rs. 7,000 + 3,600	Centrifugal fans give better static pressure through ducting/filter than axial fans
PTC ceramic heater	Controlled drying	~Rs. 2,500	Self-limiting temperature (PTC) is inherently safer than a resistive element near a chemical-processing enclosure
LCD touchscreen/encoder	UI	~Rs. 5,500	Straightforward local control without needing a phone/PC
Enclosure frame & panels	Containment	~Rs. 15,000	Core to the whole safety pitch — don't cut this budget line even under pressure
3× SMPS (24/12/5V)	Power	~Rs. 7,500	Separate rails isolate motor noise from sensor/logic power, as stated in your own safety-features slide

3.4 Approval Checklist — Hydro-Dipping Workstation

General

- ☒ Solves a real problem — strongest of the four
- ☐ Target users identified — implied (small shops, hobbyist customizers, education) but not stated on a slide; add one line
- ☒ Existing solutions researched — 10 references, good
- ☒ Improvement clearly explained — safety/monitoring layer over manual process

Hardware

- ☒ Hardware-focused
- ☒ Enough meaningful sensors — 6–7 sensors, safety-justified
- ☒ Sensors improve functionality — yes, each gates a specific unsafe condition

- ✓ Components available — mostly standard, a couple of industrial-grade items need lead time (order early)
- ✓ Low power — three regulated rails, reasonable
- ✓ Low cost — 112,200 LKR, on the higher side but justified by containment/safety hardware
- ✓ Testable in lab — yes
- ✓ Can be built — yes, though the enclosure is the most fabrication-intensive part of any of the four proposals — budget real build time for it

Proposal

- ✓ Inputs identified
- ✓ Outputs identified
- ✓ Block diagram complete
- ✓ Flowchart complete
- ✓ Component explanations ready
- ✓ Scope defined — best-specified of the four, keep the worked example in your final deck
- ✓ Budget estimated, with contingency already built in

Verdict: Approve — strongest proposal overall; fix the one weak sensor assignment (§7.2) and add a one-line target-user statement.

4. Project C — AquaScout ROV

4.1 Requirement Check

Criterion	Status	Notes
Real-world problem	⚠️ Partial	Underwater inspection accessibility is a genuine problem, but...
New/improved	⚠️ Weakest point of this proposal	Your own references (Science Buddies, Zbotic, Blue Robotics) are established DIY-ROV build guides — a supervisor will ask directly "what's different from these?" Have a sharp answer ready (see Q&A below); right now the deck doesn't state one explicitly.
Hardware-focused	✅	
AI as value-add only	✅	PID stabilization isn't really "AI," it's control theory — fine either way, just don't call it AI.
Low budget	✅	~97,750 LKR is reasonable, though the MS5837 depth sensor alone is 12,500 — over 10% of budget on one part.
Fully practical	⚠️ Partial	Practical as a demo/inspection tool at 20–50m tether range; be honest that "practical" here means shallow-water/tank inspection, not open-ocean deployment.
Testable in lab	⚠️ Needs planning	You need actual water — a tank or pool — arranged before your test phase; this isn't a bench test like the other three projects. Flag this early with your supervisor as a logistics dependency, not an afterthought.
5 members / ≥1 sensor each	⚠️ Partial	See §7.3 — one sensor (battery current) appears only in the work-distribution slide, not the architecture/hardware slides. Fix the inconsistency.
Inputs/outputs defined	✅	
Scope defined	✅	Clear 20–50m tethered scope, explicit depth range on the pressure sensor.

4.2 Supervisor Q&A Prep

Why is this better than existing products? Two honest angles, use both: (1) vs. commercial ROVs (Blue Robotics-class) — cost. Commercial units with comparable depth/telemetry capability run far above a student budget; yours targets the same capability class at a fraction of the cost using accessible parts. (2) vs. other DIY/hobbyist ROV builds — most published builds (including your own references) are single-purpose demo builds without a closed-loop stabilization system; your PID-based depth/orientation hold plus integrated leak-triggered auto-surface fail-safe is the part that's not "just another tutorial build." Lead with the PID stabilization + fail-safe combination as your novelty claim, because the hull/thruster/tether architecture itself is not novel and you should not claim it is.

Why did you choose this sensor? (see §4.3)

What happens if this sensor fails?

- Leak sensor fails/false-negative → water ingress undetected, risk to electronics and drone loss. This is your top safety risk; the deck already has leak-triggered auto-ascent — make sure the firmware also treats "no valid reading" the same as "leak detected" (fail-safe, not fail-silent).
- Depth/pressure sensor (MS5837) fails → PID stabilization loses its primary input, drone could sink or surface uncontrolled. Mitigation: define a hard "sensor timeout → cut thrusters to neutral buoyancy trim / abort dive" behavior, not left implicit in the current flow.
- IMU fails → orientation hold is lost; drone may tumble but shouldn't sink — lower severity than depth-sensor failure, say so to show you've ranked your failure modes by severity, which supervisors like to see.

Why not use another microcontroller? ESP32/Arduino Mega choice depends on which you actually finalize — be ready to justify whichever one is in the final BOM. ESP32: cheaper, Wi-Fi (not usable underwater but useful for the surface-side unit if repurposed), less native serial. Arduino Mega: more UART/PWM headroom for RS-485 + multiple sensors + 4 ESCs simultaneously without needing an I/O expander — for a system this sensor/actuator-dense, Mega's extra native serial ports are a legitimate practical reason if that's your final pick.

Estimated cost? LKR 97,750 including 10% contingency — flag the MS5837 (12,500) as your single priciest sensor and know why it's worth it: it's a marine-rated pressure sensor, not a hobby barometer, and depth accuracy is central to the whole stabilization loop.

Power consumption? Submersible side runs off 3S/4S LiPo, split into a 5V logic rail and an 11.1/14.8V motor rail via the PDB; surface side is fully decoupled (laptop + separate joystick power) — emphasize this decoupling, since it means a surface-side fault can't propagate into the submersible unit.

How will you test it?

- Dry-bench test of all electronics before any water exposure (standard ROV practice — mention this explicitly, it shows engineering discipline).
- Leak/pressure test of the sealed hull alone (no electronics) before committing the electronics sled.
- Shallow-tank test of stabilization loop (does it actually hold depth/orientation under a PID tune).
- Tethered range test at increasing distances up to the 20–50m spec, checking for RS-485 signal integrity.
- Full mission rehearsal (e.g., the pipeline-inspection scenario in your own flow slide).

4.3 Component Research

Microcontroller — ESP32 / Arduino Mega (finalize one)

Pros (ESP32): cheap, dual-core, Wi-Fi for surface-side reuse. **Cons (ESP32):** fewer native UARTs — RS-485 + sensor buses + ESC PWM simultaneously may need careful pin planning or a UART expander. **Pros (Mega):** more native serial ports, simpler for a sensor/actuator-dense build, very well documented for RS-485 shields. **Cons (Mega):** single-core, slower clock, no built-in wireless (irrelevant underwater, but relevant if you want it doubling as a surface controller too). **Cost:** ~Rs. 3,500 either way (matches budget line). **Supplier:** Standard hobby-electronics channels. **Reason to choose:** Pick Mega if RS-485 + IMU + depth + leak + camera all need simultaneous native serial/interrupt handling without an I/O expander; pick ESP32 if you're comfortable multiplexing over I2C/SPI and want the Wi-Fi option for the surface unit. **Decide and lock this before your next supervisor meeting** — the deck currently lists both as if interchangeable, which invites the "why not the other one" question you can't yet answer crisply.

Sensor 1 — MPU6050 IMU

Purpose: Orientation sensing for the PID stabilization loop (roll/pitch hold). **Accuracy:** Reasonable for stabilization (± 250 – $2000^\circ/\text{s}$ gyro range, ± 2 – 16g accel), but has known drift over time without a magnetometer/complementary filter — a fair thing to flag proactively as a limitation you've planned to filter for. **Cost:** ~Rs. 850. **Alternatives:** MPU9250 (adds magnetometer for heading reference, ~Rs. 1,200–1,800, better long-term heading stability); BNO055 (onboard sensor fusion, much less firmware work, notably pricier, ~Rs. 3,000+). **Problems:** Drift without fusion/calibration; needs a solid vibration-isolated mount since thruster vibration will inject noise into the accelerometer readings.

Sensor 2 — MS5837 Depth/Pressure Sensor

Purpose: Primary input for depth-hold PID; marine-rated pressure-to-depth conversion. **Accuracy:** Marine-grade, typically resolves to a few cm of depth — genuinely appropriate given the price tag, unlike a generic barometer. **Cost:** ~Rs. 12,500 — your single most

expensive sensor across all four proposals; own that in the viva rather than getting caught off guard by the "why so expensive" question. **Alternatives:** BMP280/BME280 (cheap, ~Rs. 300-600, but not rated for submersion or the pressure range/waterproofing this needs — the price gap is because MS5837 is designed to be potted/submerged, not because it's arbitrarily premium); dedicated industrial pressure transducer (more accurate still, far pricier, overkill for shallow-water student use). **Problems:** Needs correct waterproof potting/mounting — a sensor failure here is usually a sealing failure, not an electronics failure; budget real time for waterproofing, not just "wire it up."

Sensor 3 — Leak Detection Probe

Purpose: Detects water ingress; triggers auto-ascent fail-safe. **Accuracy:** Binary detection at the probe location. **Cost:** ~Rs. 300-600 (folded into your PVC/O-ring/epoxy line — make this an explicit budget line item, it's currently uncoded separately). **Alternatives:** Multiple probes at different hull low-points (more coverage, marginal extra cost, worth doing given this triggers your emergency-surface behavior); moisture-sensing tape (simpler, less reliable). **Problems:** Single-point detection means a leak elsewhere in the hull isn't caught until water reaches the probe — consider 2 probes at different heights inside the hull.

Sensor 4 — Camera (FPV/USB)

Purpose: Live visual inspection feed to the surface GUI. **Accuracy:** N/A (imaging, not a measurement sensor) — frame this one differently in the viva: justify by resolution/latency suitability for the tether bandwidth, not "accuracy." **Cost:** ~Rs. 4,500. **Alternatives:** Raspberry Pi Camera Module (higher quality, needs a Pi onboard — adds cost/power/waterproofing complexity you likely don't want in the submersible unit); analog FPV camera + separate video transmitter (lower latency, but you're wired via tether anyway so this loses its main advantage). **Problems:** Underwater optical clarity depends heavily on water turbidity — a real-world limitation worth acknowledging rather than promising in the demo.

Sensor 5 (added in work-distribution slide only) — Battery Voltage/Current Sensor, e.g. INA219

This is the inconsistency flagged in §7.3 — it's assigned to De Silva as their sensor, but doesn't appear in the Hardware Components or Block Diagram slides. Add it properly to both, or reassign a sensor that's actually in your architecture. **Purpose:** Monitors LiPo voltage/current draw — battery health and early low-voltage warning (LiPo damage/fire risk if fully discharged). **Accuracy:** INA219 gives mA-level current resolution and good voltage accuracy — plenty for a health/warning function. **Cost:** ~Rs. 400-600. **Alternatives:** Simple voltage divider + ADC read (cheaper, current-blind); dedicated LiPo checker module (simpler UI, less integration flexibility). **Problems:** Needs correct shunt-resistor range for your motor peak current, or you lose resolution at low-load / saturate at high-load.

Actuators & Structural Components (grouped)

Component	Function	Est. Cost	Reason to choose
4× brushless thrusters + ESCs	Propulsion (2 horizontal + 2 vertical)	~Rs. 26,000 + 8,800	Waterproof brushless is standard for small ROVs; DIY bilge-pump conversion is a valid cost-cutting fallback you've already noted
MAX485 RS-485 transceivers ×2	Tether communication	~Rs. 900	RS-485 is the standard choice for reliable signaling over tens of meters where Wi-Fi can't work underwater
Acrylic hull tube + dome	Pressure housing	~Rs. 8,000	Transparent acrylic doubles as a camera window and is easier to inspect for cracks than opaque housings
PVC/3D-printed frame	Structural support	~Rs. 5,000	Cheap, easy to iterate/reprint if buoyancy trim needs adjusting
LiPo 3S/4S + PDB + fuse/kill switch	Power	~Rs. 6,800 + 1,200 + included	Standard safe power architecture for small RC/ROV builds
Joystick/controller box	Manual piloting input	~Rs. 3,200	Direct tactile control is more reliable underwater-piloting UX than an on-screen control

4.4 Approval Checklist — AquaScout ROV

General

- ☒ Solves a real problem — accessible underwater inspection, reasonable
- ☒ Target users identified — marine research/education/small inspection, stated in intro
- ☒ Existing solutions researched — good reference list, but...
- ☐ Improvement clearly explained — **weakest item:** novelty vs. your own cited DIY-ROV guides isn't explicitly argued anywhere in the deck. Add one slide/line making the PID-stabilization + integrated fail-safe the explicit "what's new" claim.

Hardware

- ☒ Hardware-focused
- ☒ Enough meaningful sensors — yes, once the INA219 inconsistency (§7.3) is fixed
- ☒ Sensors improve functionality — yes, each maps to stabilization or safety
- ☒ Components available — yes, standard marine/hobby-ROV parts

- ✓ Low power — reasonable for a small submersible
- ✓ Low cost — 97,750 LKR, reasonable, though MS5837 is a concentrated cost risk
- ☐ Testable in lab — **needs explicit arrangement:** confirm a tank/pool is actually booked/available; this is a logistics dependency none of the other three projects have
- ✓ Can be built — yes, with real time budgeted for waterproofing/sealing, which is the highest-skill fabrication step of the four proposals

Proposal

- ✓ Inputs identified
- ✓ Outputs identified
- ✓ Block diagram complete
- ✓ Flowchart complete
- ✓ Component explanations ready
- ✓ Scope defined — clear 20-50m tether range
- ✓ Budget estimated with contingency

Verdict: Approve with revisions — fix the MCU indecision, fix the sensor-list inconsistency (§7.3), add an explicit novelty argument, and confirm pool/tank test access before your next supervisor meeting.

5. Project D — Autonomous Ring Toss Launcher Robot

5.1 Requirement Check

Criterion	Status	Notes
Real-world problem	 Weakest of the four	Ring toss is a game. The underlying tech (vision-guided precision aiming, closed-loop launch control) generalizes to real applications, but as scoped the deliverable itself doesn't solve a problem — expect this to be your supervisor's first and hardest question.
New/improved		Vision + LiDAR + closed-loop PID aiming is a real step up from a toy launcher — this is your strongest point.
Hardware-focused	 At risk	A Raspberry Pi 5 running YOLOv8 for target detection is doing a lot of the actual "intelligence" of the system — this reads as AI-centric, not AI-as-add-on.
AI as value-add only	 Direct conflict with the stated requirement	Computer vision targeting isn't a bonus feature here — remove the CV path and the robot cannot aim autonomously at all. You should either reduce CV's centrality (e.g., make LiDAR-only aiming the core, with camera/YOLO as a confirmation/refinement layer) or be ready to defend AI-centrality directly.
Low budget	 Highest of the four by far	156,420 LKR, and the Raspberry Pi 5 (8GB) alone is 45,000 LKR — nearly 29% of the entire budget on one board. This will draw scrutiny given the "low budget" requirement.
Fully practical	 Partial	Practical as a demo/competition platform, not really a "product" in daily-use terms — be upfront that this is positioned as a demonstration/research platform, not a deployable product, since that's what it actually is.
Testable in lab		Needs a safe indoor space with clearance for the firing path — arrange this and mention the tripwire interlock as your safety answer.
5 members / ≥ 1 sensor each		See §7.4 — confirmed, one member (Kalansooriya) owns two sensors.
Inputs/outputs defined		

Criterion	Status	Notes
Scope defined	⚠️ Partial	"Consistently landing rings on target" needs a number — your own operational-flow example states "within 10 cm of the peg" — put that number on the scope/objectives slide itself, not buried in a flow example.

5.2 Supervisor Q&A Prep

Why is this better than existing products? Be honest that "existing products" here are toys, not a market category with commercial competitors — your comparison should be framed as "manual/toy ring-toss launchers vs. a sensor-guided one," and your genuine claim is repeatability: closed-loop PID on flywheel RPM and aim angle removes the two biggest sources of inconsistency (speed and angle) that make human or open-loop toy aiming unreliable.

Why did you choose this sensor? (see §5.3)

What happens if this sensor fails?

- Ultrasonic tripwire (safety interlock) fails → **this is your most serious question**, since it's a physical safety system around a projectile launcher. The design must fail safe: if the tripwire sensor gives no valid signal, the system should refuse to arm/fire, not default to "assume clear." Say this explicitly and unprompted — supervisors specifically probe safety-interlock failure modes on any project that launches something.
- LiDAR fails → falls back to camera-only distance estimation (less accurate) — say you've planned this graceful degradation rather than a hard stop, since it shows resilience thinking.
- Motor encoder fails → flywheel RPM feedback lost, PID loses its control variable — should trigger a "abort shot, do not fire blind" state, not continue open-loop.
- Camera/CV pipeline fails or misdetects → system should default to requiring manual mode, not attempt a blind autonomous shot.

Why not use another microcontroller? This project genuinely needs **two** compute tiers, and you should frame it that way rather than defending one chip: ESP32 for hard real-time motor/PID/safety-interlock control (deterministic, low-latency), Raspberry Pi 5 for vision/AI (needs a real OS, more RAM/CPU than any microcontroller offers for YOLOv8). A single chip can't do both well — a Pi alone isn't real-time enough for flywheel PID and safety-interlock timing, and an ESP32 alone can't run YOLOv8. This split is defensible; just be ready to say so plainly, since it's also your best answer to "why is a Pi 5 in a microcontroller project."

Estimated cost? 156,420 LKR — be upfront that the Raspberry Pi 5 is the single largest line item (45,000, ~29% of budget) and be ready with a cost-reduction fallback (a Pi 4 or Pi Zero 2 W with a lighter model, or moving to LiDAR-primary aiming with vision as a secondary confirmation only) in case the supervisor pushes on budget.

Power consumption? Three rails off the PDB (5V for Pi 5, 3.3V for ESP32/sensors, raw battery rail via ESC for motors) off a high-discharge LiPo sized for flywheel motor surge current — know that flywheel spin-up is your peak-current event and should size the fuse/battery discharge rating around that, not average draw.

How will you test it?

- Static safety test: verify the tripwire reliably halts the system before any live-fire testing begins.
- Flywheel RPM calibration: bench-test RPM-to-launch-distance relationship at a fixed angle.
- Aiming accuracy: repeated shots at a known distance, measure landing-position spread (this gives you the number your scope slide needs — see requirement check above).
- Vision/LiDAR fusion test: compare CV-estimated vs. LiDAR-measured distance across several target positions.
- Full autonomous cycle: end-to-end detect→aim→fire→telemetry, repeated N times for a consistency statistic.

5.3 Component Research

Microcontroller — ESP32 Dev Board (real-time control tier)

Pros: Deterministic-enough for PID/motor timing, cheap, dual-core lets you separate motor control from sensor polling, handles the safety-interlock logic independent of whatever the Pi is doing. **Cons:** Can't run the vision pipeline itself — hence needing the Pi as a second compute tier. **Cost:** ~Rs. 1,500 (×2 with spare, per your budget). **Supplier:** Standard hobby-electronics channels. **Reason to choose:** Right tool for hard-real-time launch/aim/safety control; keeping this separate from the Pi is good architecture, not redundancy — say so explicitly if asked "why two brains."

Companion Computer — Raspberry Pi 5 (8GB)

Pros: Enough CPU/RAM to run YOLOv8-Nano at usable frame rates, full Linux OS for OpenCV/Python development, big community support. **Cons:** By far your most expensive single component (45,000 LKR); higher power draw and heat than a microcontroller; adds a full-OS boot-time delay and a second point of software failure. **Cost:** ~Rs. 45,000.

Alternatives: Raspberry Pi 4 (cheaper, ~Rs. 20,000–25,000, sufficient for YOLOv8-Nano at a lower frame rate — a strong cost-reduction option to have ready); Google Coral/Pi + accelerator (adds cost, better inference speed, probably overkill here); offload vision

processing to a laptop over Wi-Fi instead of an onboard Pi (cuts this cost line entirely, adds latency and a wireless dependency — a real trade-off worth mentioning as an alternative you considered). **Reason to choose (if you keep it):** Only justified if you need onboard, low-latency inference; if a laptop can sit within Wi-Fi range of the launcher for the demo, that's a legitimate way to cut nearly 30% of your budget.

Sensor 1 — Camera Module 3 (12MP)

Purpose: Visual target/peg detection for autonomous aiming. **Accuracy:** Detection accuracy depends entirely on the trained YOLO model and lighting conditions, not the sensor itself — be ready to state your model's expected detection accuracy/confidence threshold, not just the camera's megapixel spec. **Cost:** ~Rs. 9,500. **Alternatives:** Cheaper USB webcam (much less expensive, lower autofocus quality, adequate if lighting is controlled); stereo camera (adds depth estimation without needing LiDAR at all — an interesting simplification to mention as "considered but rejected due to cost/complexity"). **Problems:** Vision-based detection is lighting- and contrast-dependent — plan to test/demo under controlled, consistent lighting, and say so rather than promising outdoor-robust performance you haven't verified.

Sensor 2 — TF-Luna LiDAR

Purpose: Direct distance-to-target measurement, feeding the trajectory calculation. **Accuracy:** ToF LiDAR of this class is typically accurate to a few cm over its 0.2–8m range — genuinely reliable compared to vision-based distance estimation, which is the strongest technical argument for including it alongside the camera. **Cost:** ~Rs. 7,000. **Alternatives:** Ultrasonic rangefinder (much cheaper, ~Rs. 400, far less accurate and narrower usable range); stereo-vision distance estimate (removes the LiDAR line entirely, less reliable, more compute). **Problems:** Single-point LiDAR needs to be aimed precisely at the target, not just "in the general direction" — pairing with the camera for target-centering before taking the LiDAR reading is the right design, make sure your flow actually does this in that order.

Sensor 3 — IR Break-Beam Sensor

Purpose: Confirms the ring has actually left the feeder/launcher (feed/fire confirmation). **Accuracy:** Binary but highly reliable at its trigger point — no real accuracy concern here. **Cost:** ~Rs. 500. **Alternatives:** Microswitch on the feed mechanism (cheaper, more mechanical wear); optical encoder on the feeder servo (more precise but unnecessary complexity for a binary confirmation). **Problems:** Needs to be positioned so a ring reliably breaks the beam without false triggers from vibration.

Sensor 4 — Motor Encoders ×2 (Flywheel RPM Feedback)

Purpose: Closed-loop RPM feedback for the PID controller driving launch speed — this is the sensor most responsible for your "consistent accuracy" claim. **Accuracy:** Magnetic/optical encoders typically give fine enough resolution for stable RPM PID control; good encoder choice here directly determines shot-to-shot consistency. **Cost:** ~Rs. 3,000

for 2. **Alternatives:** Back-EMF sensing on the brushless motor (no extra sensor cost, less accurate/noisier); a single shared encoder if both flywheels are mechanically linked (cheaper, but loses independent control if you want asymmetric spin for curve/side correction — probably not worth the savings here). **Problems:** Encoder mounting/coupling slack introduces reading error — a common practical failure point to mention if asked about testing.

Sensor 5 — Ultrasonic Sensor, HC-SR04 (Safety Tripwire)

Purpose: Safety interlock — disables launch if a person/object enters the firing path.

Accuracy: Adequate for a presence/absence decision in the 2-400cm range; not the limiting factor in this safety system — the limiting factor is your failure-mode handling (see Q&A above). **Cost:** ~Rs. 400. **Alternatives:** IR break-beam across the firing path (simpler, cheaper, single-line-of-sight only); a physical light curtain (industrial-grade safety component, far more expensive and reliable — worth mentioning as the "gold standard" you're approximating on a student budget). **Problems:** Ultrasonic beams have a cone width, not a hard line — someone could be at the edge of the danger zone without triggering it. Be upfront about this limitation rather than over-claiming the safety coverage, and consider dual sensors covering the width of the firing path.

Actuators & Structural Components (grouped)

Component	Function	Est. Cost	Reason to choose
2× brushless outrunner motors + 30A ESCs	Flywheel launch	~Rs. 9,000 + 6,000	Standard high-RPM, high-torque combo for consistent launch speed
2× NEMA17 + stepper drivers	Pan-tilt aiming	~Rs. 7,000 + 2,500	Precise angular positioning for the aiming platform
MG996R servo	Ring feed mechanism	~Rs. 1,800	High torque-for-cost, reliable for a simple push-feed action
LiPo 4S ≥5000mAh	Power	~Rs. 9,000	Sized for flywheel surge current, standard RC-class battery
Power distribution PCB + fuse/E-stop	Power safety	~Rs. 4,000 + 1,500	Centralizes protection and gives a hard manual kill switch, essential for a launching mechanism
3D-printed/acrylic chassis	Structure	~Rs. 15,000 + 8,000	Low-cost, quick to iterate mechanically during tuning

5.4 Approval Checklist — Ring Toss Launcher

General

- ☐ Solves a real problem — **fails as scoped**; reframe the pitch around the transferable technology (vision-guided precision aiming/launch control), or add a genuine secondary application (e.g., an accessibility-aimed target-assist device, or a training/rehab tool) if you want this to clear a strict "real problem" bar.
- ☐ Target users identified — currently framed as "demonstration, competition, and research" — that's an audience for a project, not "users" of a product; needs work if the requirement is read strictly.
- ☒ Existing solutions researched — good reference list
- ☒ Improvement clearly explained — vision+LiDAR+PID over toy launchers is well argued

Hardware

- ☐ Hardware-focused — **at risk**: too much of the system's core function lives in the CV/AI pipeline rather than physical hardware
- ☒ Enough meaningful sensors — 6 sensors, good coverage
- ☒ Sensors improve functionality — yes
- ☒ Components available — yes, though Pi 5 supply/price can fluctuate, worth checking availability before locking budget
- ☐ Low power — Pi 5 + flywheel motors is your highest peak-power draw of the four projects; not disqualifying, but be ready with a number
- ☐ Low cost — **fails as scoped**; highest budget of the four by a wide margin, driven almost entirely by the Pi 5
- ☒ Testable in lab — yes, with a safe firing-range space and the tripwire in place
- ☒ Can be built — yes, mechanically straightforward relative to AquaScout's waterproofing challenge

Proposal

- ☒ Inputs identified
- ☒ Outputs identified
- ☒ Block diagram complete
- ☒ Flowchart complete
- ☒ Component explanations ready
- ☐ Scope defined — put the "10cm accuracy" target from your own flow example onto the actual objectives/scope slide as a measurable success criterion
- ☒ Budget estimated with contingency

Verdict: Major revisions needed before approval — this is the proposal most likely to get pushback in its current form. The three things to fix before you present it: (1) either reduce AI/CV centrality or have a direct, confident defense of the two-compute-tier architecture ready, (2) cut cost — evaluate Pi 4 or laptop-offload seriously, (3) put a numeric accuracy target on the scope slide. The underlying engineering is genuinely good; the positioning needs to catch up to it.

6. Cross-Project Comparison & Recommendation

	3D Potter Printer	Hydro- Dipping	AquaScout	Ring Toss
Real-world problem strength	Medium	High	Medium	Low
Novelty strength	Medium-High (dual mode)	Medium (safety layer)	Low-Medium (needs an explicit argument)	High (vision+PID aiming)
Budget (LKR)	~87-95k	~112k	~98k	~156k
Biggest single risk	Missing competitor slide	One weak sensor assignment	MCU undecided + novelty argument missing	AI-centrality + cost + "is this a real problem"
Test logistics	Bench only	Bench only	Needs a tank/pool	Needs a safe firing range
Fabrication difficulty	Medium (frame + extrusion)	Medium-High (enclosure)	High (waterproofing)	Medium

If you can only push hard on one proposal: Hydro-Dipping Workstation is currently your safest choice against every stated requirement — the "real problem" and "safety engineering" arguments are already strong and mostly just need polish, not rework.

If your supervisor has said they like sensor-rich projects and you want to maximize that: the 3D Potter Printer and Hydro-Dipping decks both already hit 6+ sensors with clear individual justification; Ring Toss also has 6 but carries the AI-centrality risk that could undercut the "hardware-focused" requirement regardless of sensor count.

Ring Toss needs the most work before it's presentable — not because the engineering is weak (it's arguably your most technically interesting build) but because three specific requirement lines fail as currently scoped (real-world problem, AI-as-add-on, low budget).

All three are fixable with framing and one hardware substitution (Pi 5 → Pi 4 or laptop-offload), not a redesign.

7. Work Distribution Fairness — All Four Proposals

You asked specifically whether the work distributions are fair, and noted some proposals seem to have problems. Here's the breakdown, project by project, then the overall pattern.

7.1 3D Potter Printer

#	Member	Hardware Module	Sensor
1	OMESH N.A.N	Mixing chamber, mixing motor+shaft, water pump	Moisture sensor
2	GOONATILLAKA W.A.D.S.S	Auger/piston extrusion, nozzle, clay flow	Load cell + HX711
3	DE SILVA P.S.S	LCD/TFT, keypad/encoder, SD card	Motor current sensing
4	CHAMIDYA J.A.S	MCU system, power distribution & wiring	Temp & humidity (DHT22)
5	KALANSOORIYA O.K	X/Y/Z mechanics, steppers, frame	Limit switches

Note: this mapping is inferred from slide position order (names listed 1–5, roles listed 1–5 immediately after) — the slide doesn't explicitly draw a line from each name to each role block. **Double-check this against the actual deck layout before presenting** — position-based mapping is easy to accidentally scramble during editing, and you don't want to misstate your own team's roles live.

Fairness assessment: Reasonably even. "Control System & Coordination Lead" (whoever that maps to) is somewhat lighter on hands-on hardware building (mostly wiring/integration) compared to "Motion System Lead" or "Extrusion System Lead," who each own a mechanically complex subsystem. If that person is one of your two ITM (IT & Management) members, that's a sensible pairing — coordination/integration role suits someone building toward a more systems/PM-oriented skill set, while the IT-major members take the more hands-on mechanical/electronics-heavy roles. Worth confirming that's intentional rather than accidental.

7.2 Hydro-Dipping Workstation

Member	Sensor	Hardware Module	Assessment
CHAMIDYA J.A.S (IT)	Water Temperature (PT100+MAX31865)	Water Stabilization & Heating Logic	Solid, single clear sensor + module
DE SILVA P.S.S (IT)	Water Level & Leak Detection (2 sensors)	Tank & Leak Monitoring	Heavier sensor load (2), but a coherent module
OMESH N.A.N (ITM)	VOC/Gas & Airflow (2 sensors)	Ventilation & Filtration + IoT dashboard	Heaviest overall (2 sensors + a software dashboard task)
KALANSOORIYA O.K (IT)	Position Limit Switches	Motion Control (rail, stepper, driver)	Solid, single clear sensor + module
GOONATILLAKA W.A.D.S.S (ITM)	"User Input Interface Sensing"	Spray & UI System	Weakest assignment — flagged in §3.3

Fairness assessment: Uneven in two ways. First, two members (De Silva, Omesh) own two sensors each while two others (Chamidya, Kalansooriya) own one — not drastically unfair, but worth balancing if your supervisor counts sensors per person literally. Second and more importantly, Goonatillaka's "sensor" isn't a real physical sensor, which is a genuine gap against the "everyone handles at least one sensor" rule, not just a fairness issue. **Fix:** give Goonatillaka co-ownership of a second leak sensor, or make them responsible for adding a redundant safety float switch in the drip tray (a real gap identified in §3.3 anyway) — this solves two problems at once.

7.3 AquaScout ROV

Member	Sensor	Hardware Modules	Assessment
CHAMIDYA J.A.S	Leak Detection	Hull, frame, buoyancy/ballast, cable penetrators	Heavy mechanical/waterproofing load — hardest fabrication skill on the team
DE SILVA P.S.S	Battery Voltage/Current (INA219)	LiPo, PDB, ESCs, thrusters, fuse/kill switch	Sensor not in the architecture/hardware slides — see below
OMESH N.A.N	IMU (MPU6050)	MCU, firmware, PID loop	Software-heaviest role — appropriate given firmware focus
KALANSOORIYA O.K	Camera (FPV/USB)	RS-485 modules, tether, strain relief	Solid, clear
GOONATILLAKA W.A.D.S.S	Depth/Pressure (MS5837)	Joystick, surface Arduino/laptop GUI	Solid, clear

Fairness assessment: Reasonably even in workload, but there's a real consistency bug: De Silva's assigned sensor (battery voltage/current monitor, INA219) appears **only** on this work-distribution slide — it's not in the Hardware Components list (slide 9) or the Block Diagram (slide 5). Either this sensor was added specifically to make sure all 5 members had one (which is honest but needs to be reflected everywhere, not just here), or it's a genuine oversight. Either way: add the INA219 to your hardware BOM and system diagram before your next review, or your supervisor will notice the mismatch immediately when cross-checking slides — that's exactly the kind of inconsistency a strict reviewer catches fast.

Also note Chamidya's load here (hull sealing + buoyancy + waterproofing) is the single hardest fabrication skill across all sixteen roles in this document — make sure that's a deliberate choice (their interest/strength) and not just "whoever was left."

7.4 Ring Toss Launcher

Member	Sensor(s)	Hardware Modules	Assessment
Chamidya J.A.S	IR Ring-Exit Sensor	Flywheel housing & feeder mechanism	Solid, clear
De Silva P.S.S	Voltage & current monitor	PDB, battery, fuse/E-stop	Solid, clear, safety-relevant
Omesh N.A.N	Motor encoders (RPM feedback)	ESP32 board, ESC & motor driver wiring	Solid, appropriately paired with firmware role
Kalansooriya O.K	Camera + TF-Luna LiDAR (2 sensors)	Raspberry Pi 5, pan-tilt gimbal	Heaviest role — owns the entire CV/AI stack plus 2 sensors plus the priciest single component (the Pi)
Goonatillaka W.A.D.S.S	Ultrasonic safety tripwire	Dashboard, telemetry	Lightest hardware load of the five

Fairness assessment: This is your least even distribution of the four. Kalansooriya is carrying the computer-vision pipeline (arguably the hardest software task in any of the four projects), the Pi 5 hardware integration, two sensors, and the gimbal mechanics — while Goonatillaka's role (dashboard + telemetry + one sensor) is comparatively light, especially since dashboard work is largely software/UI rather than hardware. Given the project's own "AI as value-add only" and "every member has hardware responsibilities" requirements, this imbalance compounds two separate risks at once: (1) it's the clearest sign that CV/AI has become a load-bearing part of the project rather than an add-on, and (2) one member is doing disproportionately more. **Fix options:** split gimbal mechanics off to Goonatillaka (pairing naturally with dashboard/telemetry, since the gimbal reports position back to that dashboard anyway) and have Kalansooriya focus purely on the CV/Pi stack, or move LiDAR ownership to Omesh (it's a firmware-adjacent sensor since the ESP32 likely reads it directly) to lighten Kalansooriya's sensor count from 2 to 1.

7.5 Overall Pattern Across All Four

A repeating issue: two of the four proposals (Hydro-Dipping, AquaScout) show a small inconsistency where a "sensor" was added to make the 5-way assignment work, without fully threading it back through the architecture/hardware slides. Before your next review, do one pass across **every** deck checking that the work-distribution slide's sensor list exactly

matches the hardware-components slide and the block diagram — three lists, one truth. That single fix removes two of your most easily-caught inconsistencies.

On your question about dividing roles before everyone has the required skills: that's a completely normal and sound approach for a project like this — **do not wait until everyone feels ready to divide work**. A workable sequence:

1. **Lock roles now**, based on interest and rough strength (mechanical/electronics/firmware/UI), not current mastery.
2. **Book 2-4 weeks at the start of the timeline** (this already roughly matches "Research & Requirements"/M1 in your own timelines) as a shared skill-building phase — everyone does a small guided exercise in the others' domains too (e.g., every member wires and reads one sensor on a breadboard, regardless of their assigned role), so no one is completely blind to the rest of the system when integration time comes.
3. **Pair the heavier/harder roles with a check-in buddy** — e.g., in AquaScout, Chamidya's waterproofing work and Kalansooriya's CV work in Ring Toss are both higher-risk, higher-skill tasks; assign a second member as a standing second-opinion/testing partner for those two roles specifically, without changing primary ownership.
4. **Rebalance once, at roughly the 1/3 mark**, once real skill levels and interests have shown themselves — it's fine and expected that the initial division shifts slightly once people know more about what they're actually good at or enjoy.

This gives you a clear answer for a supervisor who asks "how did you decide who does what" — you can honestly say roles were assigned by initial interest/strength, backed by a shared foundational skill-building phase, with a planned rebalancing checkpoint.